ModelIt!

Teacher's Guide

What Happens When You Push Harder?

Force, Speed, and How Things Move

Quick Reference

Grade Level	Kindergarten
NGSS Standard	K-PS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: K-PS2-1

Plan and conduct an investigation to compare the effects of different strengths or different directions of pushes and pulls on the motion of an object.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students plan and conduct a push strength investigation, comparing gentle versus strong pushes on the same ball and observing speed differences.
Disciplinary Core Idea	**PS2.A Forces and Motion** — Pushes and pulls can have different strengths and directions. A bigger push makes an object move faster than a smaller push applied to the same object.
Crosscutting Concept	**Cause and Effect** — Students observe that changing push force (cause) directly changes ball speed (effect) — a stronger push produces a faster ball.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The relationship between force and motion is one of the most fundamental principles in physics. Understanding that bigger forces produce greater changes in motion helps students make sense of everyday experiences — from kicking a soccer ball to pushing a friend on a swing.

Force and Speed

Force is any push or pull that acts on an object. When a force is applied to a stationary object, it begins to move. The greater the force, the faster the object moves. This relationship is described by Newton's Second Law of Motion: Force equals mass times acceleration ($F = ma$). At the kindergarten level, this translates to a simple, observable truth — push harder, and things go faster. A gentle finger tap on a ball gives it very little speed. A strong two-handed push makes the same ball zoom across the room.

Comparing Different Forces

Scientists compare forces by changing one variable at a time. In this lesson, students keep the ball the same but change the push strength. This controlled investigation reveals a clear pattern: gentle tap = slow, medium push = medium speed, strong push = fast. This pattern holds true for all objects — the same force principle applies whether you are pushing a ball, swinging a bat, or pedaling a bicycle. By age 5-6, children have extensive intuitive experience with this concept; the lesson helps them articulate and formalize what they already know from play.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Push Force as the external component (we choose how hard to push) and Ball Speed as the internal component (the ball's speed changes based on how hard we pushed it).

E **Establish Relationships**
Students discover that more Push Force leads to faster Ball Speed — a positive relationship.

V **Visualize & Model**
Students build a simple two-part model connecting Push Force to Ball Speed with a positive arrow.

E **Evaluate Outcomes**
Students test Gentle Tap vs. Power Push and observe the clear difference in how fast the ball rolls.

R **Revise & Extend**
Students explore what happens when they push different types of balls (heavy basketball vs. light tennis ball) with the same force — does the heavy ball go as fast?

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Watch this! (Gently tap a ball.) Now watch THIS! (Push it hard.) What was different? Today we find out WHY harder pushes make things go faster!"*

Watch For

Demonstrate two pushes on the same ball — a tiny tap and a big push. Students describe what they saw. Ask: Which was faster? WHY?

Slide 2: What Will We Learn?

3 min

Say: *"Today we learn about FORCE and SPEED! Force is how hard you push. Speed is how fast things go. More force means more speed!"*

Watch For

Introduce 'speed' and 'force.' Practice: tiptoe slowly (slow speed), run in place fast (fast speed). Gentle clap (small force), big clap (big force).

Slide 3: Let's Be Scientists!

3 min

Say: *"Our big question is: What happens when you push a ball HARDER? Does it go faster, slower, or the same? Let us investigate!"*

Watch For

Quick prediction vote: hands up for faster, hands down for slower, hands sideways for the same. Most students will correctly predict faster!

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts. Push Force is how hard we push — WE choose that! Ball Speed is how fast the ball goes — that depends on our push."*

Watch For

Use sorting mats. Students place Push Force on OUTSIDE and Ball Speed on INSIDE. Practice: show a gentle push motion and a power push motion.

Slide 5: How Are They Connected?

5 min

Say: *"When Push Force goes UP, Ball Speed goes UP! That is a POSITIVE connection. Harder push means faster ball!"*

Watch For

Line up students for a push test. Each student pushes the same ball gently, then hard. The class observes: Was it faster the second time? YES! Thumbs up for positive.

Slide 6: What Did We Find Out?

5 min

Say: *"We discovered the secret to speed — FORCE! A bigger push gives the ball more speed. A tiny push gives only a tiny bit of speed. This works for EVERYTHING!"*

Watch For

Chant: 'When Push Force goes up, Ball Speed goes up!' Complete sentence frame. Students show force with body — gentle push gesture, then POWER push gesture.

Slide 7: STEM Challenge

10 min

Say: *"Time for Speed Bowling! Line up the pins and find the PERFECT push. Too gentle and the ball stops short. Too hard and it goes too fast. Find the sweet spot!"*

Watch For

Set up plastic bottle pins. Students take turns pushing a ball at the pins. Try gentle, medium, and strong pushes. Record which push worked best using smiley faces. Crown a Speed Bowling Champion!

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Push Force
- **INTERNAL:** Ball Speed

Relationships

- **Push Force to Ball Speed:** POSITIVE (+) — When you push harder, the ball rolls faster because a bigger force transfers more energy to the ball, making it speed up more.

Key Discoveries

- A harder push makes the ball go faster — more force means more speed!
- A gentle tap makes the ball roll slowly because there is less force
- The same ball can go fast OR slow depending on how hard you push it
- Force is what changes how things move — without a push, the ball stays still

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Heavy things always go slower"	A heavy ball pushed with a strong force can go very fast. Weight does not determine speed — force does. A bowling ball rolls fast because the bowler pushes it hard. The same ball would barely move with a gentle tap. Speed depends on how much force is applied, not just how heavy the object is.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Once you push something it will go forever"	Objects slow down and stop because of friction — the rubbing between the ball and the floor. Friction is a force that works against motion. On a smooth floor, the ball rolls farther because there is less friction. On carpet, it stops sooner because there is more friction.